

- 1 In an investigation to find the water potential of onion tissue, some students used epidermal tissue from the storage leaves of a red onion. The vacuoles of the cells in this tissue contain a red pigment.

The students researched one method to use in this investigation. The first step in this method is to find the concentration of sucrose solution at which 50% of the cells in the epidermal tissue are plasmolysed and 50% are not plasmolysed.

The students decided to use sucrose solutions of different molar concentrations in their investigation and record the effect on the number of cells plasmolysed in the onion epidermis.

- (a) State the independent variable **and** the dependent variable in this investigation.

independent variable

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dependent variable

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[2]

- (b) (i) Describe how the students could make 250 cm³ of a 1.0 mol dm⁻³ solution of sucrose. (mass of one mole of sucrose molecules = 342 g)

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- (ii) Outline how the students should use **proportional** dilution to make a suitable number and range of sucrose concentrations from the 1.0 mol dm⁻³ solution of sucrose. 20 cm³ of each concentration will need to be prepared.

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- (iii) Suggest why a cell with a coloured pigment in the vacuole is suitable for the students' investigation.

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- (v) Describe how the students could use their results to find the concentration of sucrose solution in which 50% of the cells in the epidermal tissue are plasmolysed.

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- (c) The students then investigated the water potential of two other tissues that have epidermal cells with coloured pigment in their vacuoles.

For each of the three tissues investigated, the students determined the concentration of sucrose solution in which 50% of the cells were plasmolysed. All three investigations were carried out at 20 °C.

Table 1.1 shows these results.

Table 1.1

type of tissue	concentration of sucrose solution in which 50% of the cells were plasmolysed / mol dm ⁻³	water potential of the tissue / MPa
leaf epidermis of red onion	0.35	−0.85
leaf stalk epidermis of rhubarb	0.50	−1.22
petal epidermis of antirrhinum	0.44	−1.07

Use the data in Table 1.1 to explain the direction of water movement occurring in isolated cells from each of these three tissues placed in a solution with a water potential of −1.16 MPa.

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[Total: 18]